

Deforestation Trajectories and Carbon Emissions

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Objectives

The main objective of this project was to use **unsupervised learning** to group countries and subnational regions according to their **deforestation trajectories** from 2001 to 2022. Rather than focusing only on total forest loss, we wanted to identify whether different regions followed distinct temporal patterns, such as low and stable loss, sustained moderate loss, or high-intensity hotspot behavior.

A secondary objective was to extend the analysis by examining how deforestation is related to **forest carbon emissions**. After identifying meaningful loss trajectories, we asked whether forest-loss and carbon-stock variables could help explain or predict differences in country-level gross carbon emissions.

Data Preparation

Data Procurement

We downloaded the **Global Forest Data: 2001–2022** dataset from **Kaggle**. The dataset contains over **20,000 rows** representing countries, states, provinces, and territories worldwide, along with more than **20 features**, including a year-2000 forest extent baseline and **22 annual tree cover loss variables** covering the period from 2001 to 2022.

For the main unsupervised-learning workflow, we used two deforestation files from this dataset:

- a **country-level tree cover loss** file
- a **subnational tree cover loss** file

These files were selected because they directly support the project's main objective: clustering regions according to their **deforestation trajectories over time**.

Data Quality and Cleaning

The two deforestation datasets were loaded separately, inspected, and then combined into a single working dataset. We validated the structure by checking:

- duplicate full rows
- duplicate geographic-threshold keys

- missing values
- blank text values in key geographic columns
- negative values in the main numeric variables

The data quality was generally strong. The annual tree cover loss variables were complete, and the only missing values in the initial combined dataset appeared in 'subnational1', which was expected for national-level observations. These missing labels were resolved by assigning national-level rows to 'National', and a unified 'region_name' identifier was created to give each record a consistent geographic label.

Threshold Handling

A key preparation challenge was that the same geographic unit appeared multiple times under different **canopy-density thresholds**. This meant that threshold choice had to be treated as a methodological decision rather than a simple filter.

We compared thresholds **10, 30, and 50** using:

- average 2000 forest extent
- average cumulative loss
- median cumulative loss
- relative cumulative loss as a share of initial forest extent
- loss values at the beginning, middle, and end of the study period

The comparison showed that lower thresholds produce larger measured forest extents and higher absolute loss totals, while stricter thresholds reduce both. Threshold **30** was selected because it provided a reasonable middle-ground forest definition and supported a more consistent clustering workflow.

Preparing the Clustering Dataset

After selecting threshold 30, we reduced the dataset to one observation per region under a single forest-definition rule. To make deforestation trajectories comparable across regions of very different size, annual tree cover loss was normalized by each region's **forest extent in 2000**, not by total land area. This made the features represent the share of the original forest base lost each year, which is more consistent with the project objective of clustering deforestation trajectories.

Regions with zero initial forest extent were removed, and a minimum baseline forest threshold of **1,000 hectares** was applied to exclude observations with extremely small forest baselines that could create unstable percentage-based trajectories. This left **3,029 region-level observations** for clustering.

Extending the Dataset for Carbon Analysis

To support the secondary predictive analysis, we also loaded two carbon datasets from the same Kaggle source:

- a **country-level carbon** file
- a **subnational carbon** file

Their column names were standardized and aligned with the deforestation files. Structural validation showed that the main carbon outcome variables, such as gross emissions, gross removals, and net flux were fully missing at thresholds **0-25** but fully populated at thresholds **30, 50, and 75**. This confirmed that the missingness was structural rather than accidental.

For consistency with the main deforestation analysis, the carbon workflow was also restricted to **threshold 30**. At the country level, the threshold-30 deforestation and carbon datasets merged cleanly into a **236-country modeling table** with no missing values in the main analysis variables.

Tools Used for Preparation

The main preparation tools were:

1. **Pandas**, for loading, cleaning, merging, filtering, reshaping, and feature engineering
2. **Descriptive summaries and validation checks**, including missing-value tables, duplicate-key checks, threshold summaries, and numeric range checks

Challenges

Two major challenges emerged during data preparation:

1. The same region appeared under multiple threshold definitions, requiring a careful threshold comparison before clustering.
2. Percentage-based normalization became unstable for regions with extremely small initial forest extents, which required additional filtering before model fitting.

Model Evaluation — Clustering Results

1. Clustering Model: Forest Cover Threshold and Normalization

For the clustering analysis, we retained the **annual percentage of forest cover loss** for each country to identify countries with similar forest-loss trends over time. To make these trends comparable across countries, we added the **total area of each country** and normalized annual forest loss by country area. This was important because countries vary greatly in size. For example, although Brazil may have a high absolute amount of forest loss, this loss represents a smaller proportion of its total area due to its large size. Normalization therefore allows us to compare the **relative intensity of forest loss** across countries rather than relying on absolute values.

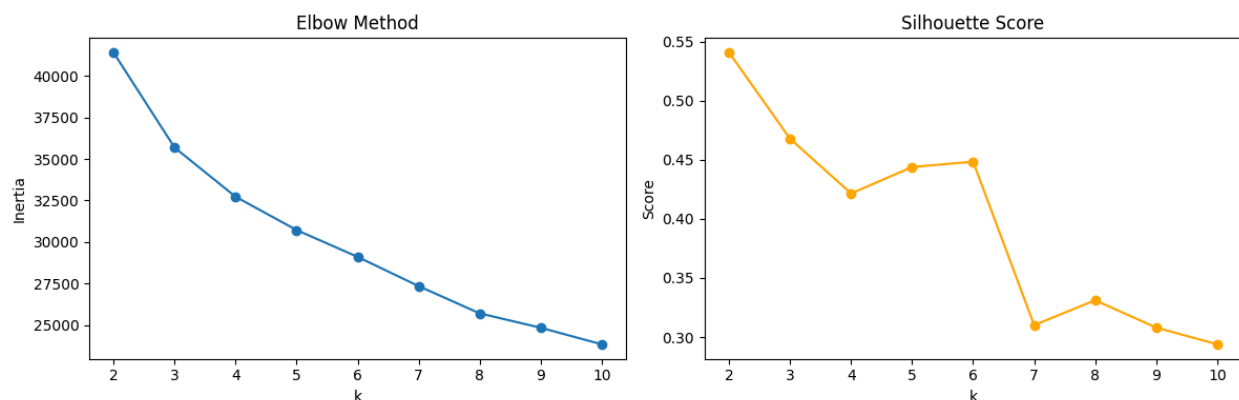
A notable pattern was already visible at this stage: **forest loss in 2016 and 2017 was substantially higher than in most other years.**

Column	Mean	Min	Max
tc_loss_ha_2001	0.098275	0.0	3.364440
tc_loss_ha_2007	0.132233	0.0	2.612491
tc_loss_ha_2012	0.157763	0.0	6.664890
c_loss_ha_2017	0.453185	0.0	60.773759
tc_loss_ha_2022	0.162576	0.0	3.133618

After normalizing annual forest loss by country area, we proceeded with the clustering analysis. First, the data were standardized using the StandardScaler and its fit_transform method to ensure that all variables were on a comparable scale.

We then applied **PCA** to reduce the dimensionality of the dataset from 21 variables while retaining 90% of the total variance. This resulted in **11 principal components**, which together explained approximately **90.8% of the variance** in the original dataset.

Following dimensionality reduction, we applied the **K-Means clustering algorithm** with `n_init = 10`. To determine the most appropriate number of clusters, we evaluated both the elbow method and the silhouette score.



The three-cluster solution divides the regions into:

- a large **Low and Mostly Stable Deforestation** group
- a **Moderate and Sustained Deforestation** group
- a small **High-Intensity Deforestation Hotspots** group

This segmentation is supported by the cluster-size distribution.

Cluster ID	Descriptive Label	Number of Observations
0	<i>High-Intensity Deforestation Hotspots</i>	622
1	<i>Low and Mostly Stable Deforestation</i>	2952
2	<i>Moderate and Sustained Deforestation</i>	162

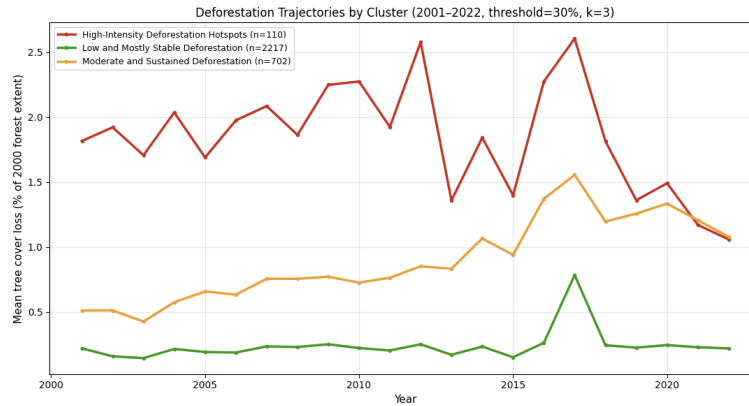
Cluster Sizes for Final K-Means Solution (k = 3)

Cluster Interpretation

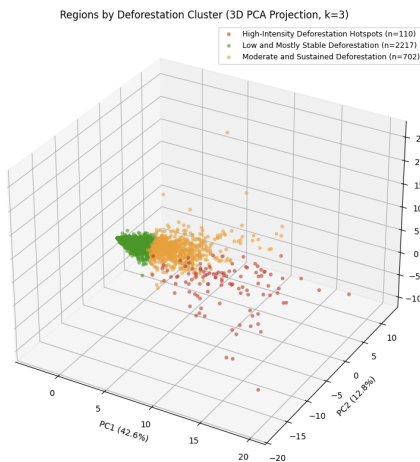
Cluster 0 (n = 622) represents a moderate forest-loss cluster. Countries in this group experience greater tree cover loss than the baseline cluster, but the magnitude and consistency of the losses are generally lower than those observed in Cluster 2. This cluster therefore represents an intermediate pattern of forest loss between the baseline and hotspot groups. Examples include countries such as the DRC and other countries showing moderate but notable levels of forest loss.

Cluster 1 (n = 2,952) can be considered the baseline or typical cluster, as it contains the largest number of observations. Countries in this cluster generally show relatively low or stable levels of tree cover loss over the study period, representing the overall pattern observed across most of the dataset. Examples of countries included in this cluster are Puerto Rico, Russia, and the Philippines.

Cluster 2 (n = 162) represents the forest-loss hotspot cluster. Countries in this group consistently exhibit higher-than-baseline tree cover loss, with particularly pronounced increases in some years. The cluster includes several tropical countries, particularly in Southeast Asia, such as Indonesia, Laos, and Vietnam. These regions have experienced substantial deforestation associated with activities such as palm oil production, logging, and the conversion of forested land to agricultural use [1]. The concentration of these countries within the hotspot cluster highlights the need for stronger forest-management policies and conservation measures in areas experiencing persistent and high rates of deforestation.

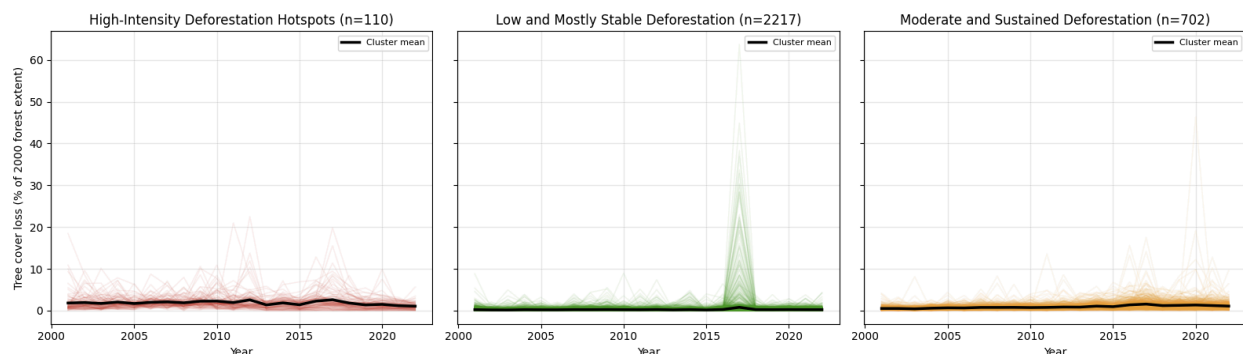


The **3D PCA projection** reinforces this interpretation by showing that the hotspot group remains visually separated from the broader low-loss baseline across multiple reduced dimensions, while the moderate cluster continues to occupy an intermediate zone.



To better understand internal cluster consistency, we also plotted the individual trajectories within each cluster together with the cluster mean. These plots show that the low-loss cluster is relatively compact overall, though it contains some observations with sharp spikes around **2017**. The moderate cluster is more elevated and sustained over time, while the hotspot cluster displays the greatest variability and the highest relative losses. This visual evidence supports the interpretation that the three groups differ not only in average intensity, but also in internal heterogeneity.

Within-Cluster Deforestation Trajectory Variation



As a robustness check, we repeated K-Means clustering on the full standardized trajectory features **without PCA**. The resulting cluster assignments showed near-perfect agreement with the PCA-based solution, indicating that PCA improved efficiency and reduced dimensionality without materially changing the underlying grouping structure.

Overall, the clustering results suggest that the dominant difference across regions is the relative intensity and stability of forest loss over time. The low-loss cluster represents the broad global baseline, the moderate cluster captures regions with sustained deforestation pressure, and the hotspot cluster identifies a much smaller set of regions with consistently higher and more volatile deforestation trajectories.

Model Evaluation — Predictive Emissions Extension

1- The three-cluster model, applied to standardized PCA-reduced data, provided a strong representation of tree cover loss trajectories from 2001 to 2022 using a **30% canopy cover threshold**. PCA reduced the data from 21 to 11 dimensions while retaining approximately **90% of the variance**.

The model identified three distinct groups. The largest cluster (79% of regions) showed consistently low annual tree cover loss of approximately 0.03–0.07%, representing the baseline pattern. The moderate-deforestation cluster showed an increasing trend, from approximately 0.2% in 2001 to 0.4–0.7% by 2022. The smallest cluster represented high-intensity deforestation, with annual losses around 20–30 times higher than the baseline, reaching approximately 1.7% of area per year. This hotspot included tropical countries such as Indonesia, Malaysia, and Laos, where deforestation is strongly associated with agricultural expansion and palm oil production. The increase in forest loss around **2016–2017** also aligns with documented global and Southeast Asian deforestation trends

2- As a complementary supervised-learning extension, we modeled country-level gross forest carbon emissions using forest-loss and carbon-stock variables. Before fitting predictive models, we compared two alternative representations of forest loss:

- absolute cumulative forest loss (cumulative_loss_ha)
- relative cumulative forest loss (cumulative_loss_pct)

The comparison showed a major difference in predictive usefulness: absolute cumulative loss was very strongly correlated with gross emissions ($r = 0.895$), while relative cumulative loss had almost no linear relationship with emissions ($r = 0.073$). This indicates that emissions are much more closely tied to the absolute scale of forest clearing than to the share of the forest base lost.

Variable	Correlation (r)
Cumulative forest loss (ha)	0.895
Cumulative forest loss (% of 2000 forest extent)	0.073

Pearson Correlation Between Alternative Forest-Loss Measures and Gross Forest Carbon Emissions

We then compared two predictive models on the same country-level emissions task:

- Linear Regression
- Random Forest Regressor

Both models performed strongly, but Random Forest was the more predictive model. It achieved the highest R^2 and the lowest MAE and RMSE, indicating better overall predictive performance.

Model	R^2	MAE	RMSE
Linear Regression	0.967	9774348	22268140
Random Forest	0.978	7085422	18383090

Table: Predictive Model Comparison for Gross Carbon Emissions

The Random Forest feature-importance ranking showed the most important predictors were:

- Cumulative_loss_ha
- carbon_stocks_2000_MgC
- Recent_loss_ha
- Early_loss_ha
- extent_2000_ha_loss

This suggests that gross carbon emissions are driven mainly by the absolute amount of forest loss and the total baseline carbon stock, while forest gain and average carbon density per hectare contribute relatively little additional predictive power in this specific model.

Rank	Feature	Importance
1	cumulative_loss_ha	0.245777
2	carbon_stocks_2000_MgC	0.217537
3	recent_loss_ha	0.174977
4	early_loss_ha	0.163507

Rank	Feature	Importance
5	extent_2000_ha_loss	0.149687
6	gain_2000-2020_ha	0.048086
7	avg_carbon_stocks_2000_MgC_ha	0.000430

Table: Random Forest Feature Importance for Gross Carbon Emissions

These results should be interpreted as explanatory rather than causal. The target variable is conceptually related to forest clearing and carbon availability, so the model is best viewed as identifying strong predictive relationships rather than discovering independent causal drivers.

Conclusion

The main unsupervised-learning workflow successfully grouped 3,029 regions into three meaningful deforestation trajectory clusters under a consistent forest-definition rule. These clusters differ mainly in the relative intensity and stability of forest loss over time, with a large low-loss baseline group, a moderate sustained-loss group, and a small but clearly distinct hotspot group. The PCA projections, cluster-mean trajectories, and within-cluster variation plots all support this interpretation, and the comparison with a non-PCA clustering solution showed that the grouping structure is robust. Also, **some** countries appeared in multiple clusters, such as Russia, Vietnam, and the Philippines. Although this initially appeared to be an issue, it is consistent with the subnational nature of the data. Different regions within the same country can have very different forest-loss patterns due to differences in climate, land use, and regulations. This is particularly relevant for large countries such as Russia, where regional forest-loss trajectories can vary considerably. A notable pattern throughout the analysis was the peak in tree cover loss around 2016–2017 across all three clusters. This increase is consistent with documented global trends, particularly in Indonesia and Southeast Asia, where deforestation increased due to palm oil expansion. Globally, increased wildfire activity also contributed to higher forest loss during this period, with El Niño-related climate conditions considered an important contributing factor [2].

The secondary supervised-learning extension showed that the relationship between deforestation and emissions depends strongly on how forest loss is represented. For clustering trajectories, **relative forest loss** was the right representation because it allowed comparison across differently sized forest regions. For predicting gross carbon emissions, however, **absolute forest loss** was far more informative, which is consistent with the fact that emissions are also measured in absolute terms.

Finally, the emissions modeling comparison showed that **Random Forest** outperformed **Linear Regression**, although both models performed strongly. This suggests that emissions are highly structured by a combination of absolute forest-loss scale and baseline carbon stocks, and that a flexible non-linear model captures these relationships slightly better than a simple linear model.

References

- 1- <https://www.sustainablepalmoilchoice.eu/deforestation-palm-oil/>
- 2- <https://www.theguardian.com/environment/ng-interactive/2018/jun/27/one-football-pitch-of-forest-lost-every-second-in-2017-data-reveals>